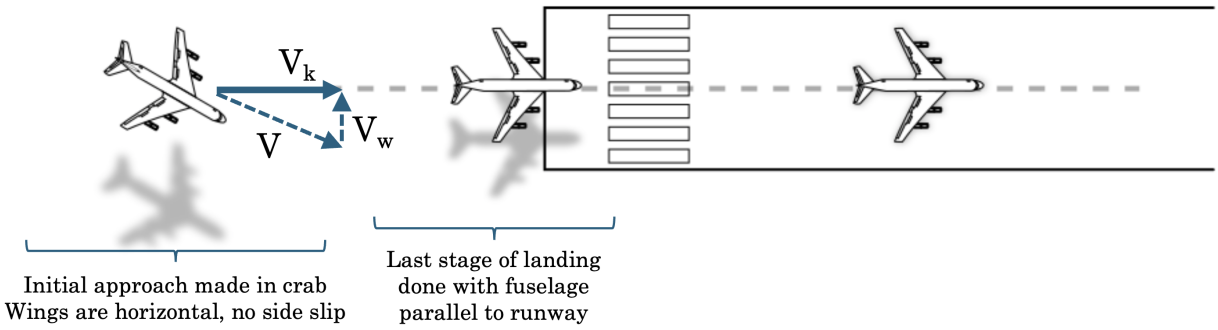
M&AE 4070/5070 - Flight Dynamics

Case Study 6 SOLUTIONS

Landing with significant cross-wind

This case study investigates landing for a typical transport aircraft when there is some cross-wind, and the limitations associated with it. The specific scenario considered here is illustrated in the figure below:



The initial approach to the airport is made in "crab", that is, with the fuselage axis aligned with the aerodynamic velocity (and therefore, at an angle compared to V_k, the velocity with respect to the ground). In that crab configuration, there is no sideslip, and the wings are horizontal. Staying in crab would however become tricky when touching down, as the wheels would not be aligned with the runway axis. To prevent landing gear strain, pilots therefore "decrab" before touchdown using lateral controls, aligning the nose with the runway and transitioning to a non-zero sideslip. This case study focuses on that last part of the landing and treats it as a steady-state flight configuration. You are tasked with determining how the lateral controls need to be adjusted, and identifying the limitations associated with such a landing.

The case study has two parts:

- In the first part, you will establish the equations governing steady flight during the "decrab" part of the landing.
- In the second part, you will calculate the position of the lateral controls and attitude of the aircraft before touching down, and the maximum cross-wind at which an aircraft can land.

1 Appropriate set of lateral equations

The first step is to simplify the steady equations for lateral motion for the specific configuration of interest. There are 2 equations coming from angular moment balance, one from the lateral force balance, and one kinematic equation:

- **R** - Equation for the roll moment, around $\overrightarrow{x_b}$:

$$C_\ell = C_{\ell_\beta}\beta + C_{\ell_p}\frac{p\ell}{V} + C_{\ell_r}\frac{r\ell}{V} + C_{\ell_{\delta_\ell}}\delta_\ell + C_{\ell_{\delta_n}}\delta_n = 0 \tag{1}$$

- **Y** - Equation for the yaw moment, around $\overrightarrow{z_b}$:

$$C_{n_=}C_{n_\beta}\beta + C_{n_p}\frac{p\ell}{V} + C_{n_r}\frac{r\ell}{V} + C_{n_{\delta_\ell}}\delta_\ell + C_{n_{\delta_n}}\delta_n = 0 \tag{2}$$

- **LF** - Balance of forces along $\overrightarrow{y_b}$:

$$mV\left(r\cos\alpha_e - p\sin\alpha_e\right) = mg\sin\phi\cos\theta_e + \frac{1}{2}\rho SV^2\left(C_{Y_\beta}\beta + C_{Y_{\delta_n}}\delta_n\right) \tag{3}$$

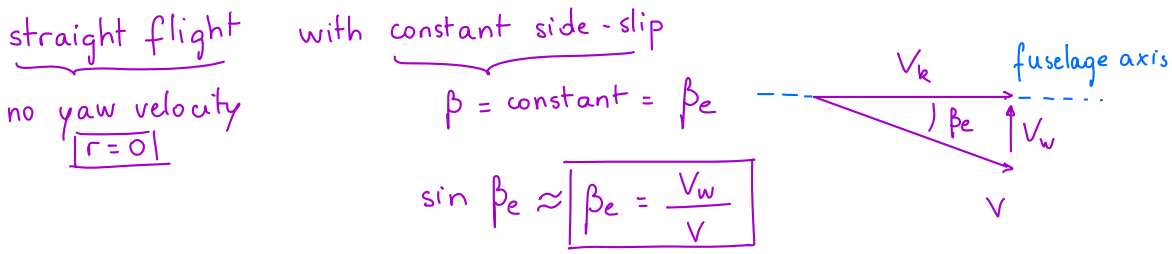
- **Kϕ** - Kinematic equation for the roll attitude ϕ around $\overrightarrow{x_b}$:

$$p = -\tan\theta_e(r\cos\phi + q_e\sin\phi) \tag{4}$$

Since there are 4 state variables (roll velocity p, yaw velocity r, side-slip angle β, and the roll/bank angle ϕ) and two additional control variable (δ_ℓ and δ_n), we need another 2 equations to close the system.

1. What are the additional two equations for our specific scenario (steady straight flight with constant side-slip)? Express those conditions in terms of aerodynamic and wind velocity magnitudes.

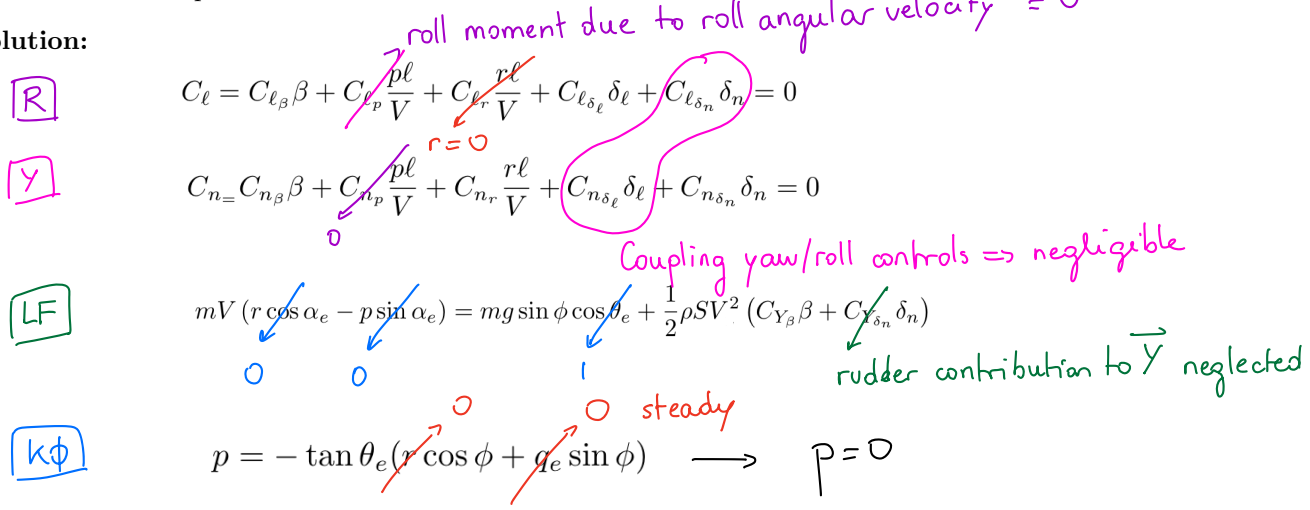
   Solution:



2. Simplify the steady equations using the following assumptions:

   - The yaw and roll moments due to roll velocity are assumed to be very small.
   - Coupling between yaw and roll controls are negligible (i.e. aileron deflection does not significantly contribute to the yaw moment, rudder deflection does not significantly contributes to the roll moment.
   - Neglect the contribution of the rudder to the lateral force.
   - The aircraft's pitch is small.

   Solution:



Simplified equations:

$$\boxed{R} \quad C_\ell = 0 = C_{\ell_\beta}\,\beta + C_{\ell_{\delta_\ell}}\,\delta_\ell$$

$$\boxed{Y} \quad C_n = 0 = C_{n_\beta}\,\beta + C_{n_{\delta_n}}\,\delta_n$$

$$\boxed{LF} \quad mg\,\phi + \frac{1}{2}\rho S V^2 C_{Y_\beta}\,\beta = 0$$

$$+ \quad r = p = 0\,, \quad \beta = \beta_e$$

6 variables $(\beta, \phi, r, p, \delta_\ell, \delta_n)$ and 6 equations

3. For each equation, explain what each term is and identify the physical role of the controls in maintaining a steady flight.

Those equations show that the ailerons and rudder deflections are both proportional to the constant side slip β_e

- to that side-slip creates a roll moment (mostly due to dihedral effect C_{ℓ_β}) which is balanced by the ailerons deflection δ_ℓ
- It also creates a yaw moment (from C_{n_β}, weather-vane effect from vertical stabilizer), balanced by the rudder deflection δ_n
- The banking, by introducing a component of the weight onto the $\vec{y}_b$ axis, balances the lateral force due to the side slip.

4. The approach is done at constant angle of attack. Express the bank angle as a function of the cross-wind velocity V_w. When the cross-wind comes from the right of the pilot, should they bring the right or the left wing down?

Approach is done at constant α.
Since the $\boxed{S}$ equation must hold too, $\boxed{LF}$ can be rewritten:

$$\boxed{S} \quad mg = \frac{1}{2}\rho S V^2 C_z$$

$$\Rightarrow \boxed{LF} \quad \sin\phi = -\frac{C_{Y_\beta}}{C_z}\,\beta$$

ϕ is assumed to be small enough and $\sin\beta \approx \beta = \frac{V_w}{V}$

$$\sin\phi \approx \phi = -\frac{C_{Y_\beta}\,\beta}{C_z} = -\frac{C_{Y_\beta}}{C_z}\,\frac{V_w}{V} = \boxed{-\sqrt{\frac{S}{mg}}\sqrt{\frac{1/2\,\rho}{C_z}}\,C_{Y_\beta}\,V_w = \phi}$$

If cross wind comes from the right, $\beta > 0$ (fuselage to the left of aero velocity vector), and $C_{Y_\beta} < 0 \Rightarrow \phi > 0$: RIGHT WING DOWN

5. Consider the aircraft data provided below.

Coefficients	C_Y	C_ℓ	C_n
β	$C_{Y_\beta} = -1.5$	$C_{\ell_\beta} = -1.3$	$C_{n_\beta} = 1.75$
r	$C_{Y_r} = 0$	$C_{\ell_r} = 2.9$	$C_{n_r} = -7.5$
δ_ℓ	$C_{Y_{\delta_\ell}} = 0$	$C_{\ell_{\delta_\ell}} = -0.33$	$C_{n_{\delta_\ell}} = -0.125$
δ_n	$C_{Y_{\delta_n}} = 0$	$C_{\ell_{\delta_n}} = 0.25$	$C_{n_{\delta_n}} = -1$

In addition,

- $\ell = 6.6$ m
- $m = 1.2 \times 10^5$ kg
- $S = 260$ m^2

(a) **Bank angle** To avoid the engines or the wings touching the ground, the aircraft cannot go above a bank angle $\phi_{\max} = 0.25$ rad $= 14^o$. Calculate the maximum allowable crosswind in the case of an approach at $V = 60$ m/s ≈ 120 kt.

Solution: The side-slip resulting from this landing configuration is equal to:

$$\beta = \frac{V_w}{V} \tag{5}$$

LF shows that the aircraft must bank to compensate with the component of the weight on the wings the lateral force created by the side-slip.

$$\phi = -\frac{\frac{1}{2}\rho S C_{Y_\beta}}{mg}\frac{V_w}{V}V^2 = 1.95 \times 10^{-4} V V_w = 1.95 \times 10^{-4} V^2 \beta \tag{6}$$

For an approach at 60 m/s, we obtain $\phi = 0.012 V_w$.
The maximum crosswind that can be handled is

$$V_{w,\max} = \frac{\phi_{\max}}{0.012} = \frac{0.25}{0.012} = 21 m/s = 40 kt \tag{7}$$

In practice, a classical aircraft is limited to 30 kt of crosswind, so this first approximation is realistic.

(b) **Control deflection** Consider a crosswind of $V_w = 30$ kt, calculate the position of the lateral controls δ_ℓ and δ_n during the approach. Are those values acceptable? How would those deflections change if you are not neglecting $C_{\ell_{\delta_n}}$?

Solution: The deflection of the controls to balance this straight flight with side-slip is obtained from the **R** and **Y** equations. Let's assume first that the cross coefficients can be neglected: $C_{\ell_{\delta_n}} = C_{n_{\delta_\ell}} = 0$. In this case, the roll control δ_ℓ is used to counter the effect of the side-slip on the roll moment (dihedral effect)

$$\delta_\ell = -\frac{C_{\ell_\beta}}{C_{\ell_{\delta_\ell}}}\beta = -3.94\beta \tag{8}$$

$$\delta_n = -\frac{C_{n_\beta}}{C_{n_{\delta_n}}}\beta = 1.75\beta \tag{9}$$

$$\beta_{\max} = \frac{30kt}{120kt} = 0.25 rad = 14^o \tag{10}$$

leading to

$$\delta_\ell = -56^\text{o} \tag{11}$$
$$\delta_n = \ \ 24.5^\text{o} \tag{12}$$

The value for δ_ℓ is not aceptable in practice, the simplifying assumptions are too brutal. We need to at least account for $C_{\ell_{\delta_n}}$, going back to $\boxed{\mathbf{R}}$:

$$C_{\ell_{\delta_\ell}}\delta_\ell + C_{\ell_{\delta_n}}\delta_n = -C_{\ell_\beta}\beta \tag{13}$$
$$C_{\ell_{\delta_\ell}}\delta_\ell = \left(-C_{\ell_\beta} - C_{\ell_{\delta_n}}\frac{-C_{n_\beta}}{C_{n_{\delta_n}}}\right)\beta \tag{14}$$
$$C_{\ell_{\delta_\ell}}\delta_\ell = (1.3 - 0.25 \times 1.75)\beta = 0.86\beta \tag{15}$$
$$\delta_\ell = \frac{0.86}{-0.33} = -2.6\beta = -36^\text{o} \tag{16}$$

This time, δ_ℓ is acceptable, even though it is very high and at the efficiency limit. This case is therefore a dimensioning case for the design of the ailerons. Note that if the dihedral effect was smaller, this control would play a smaller role.

6. In general, are fast aircrafts more or less sensitive to cross-winds at landing compared to slower aircrafts? Why?

Solution: The approach is done at constant C_z. The $\boxed{\mathbf{S}}$ equation then gives us:

$$V^2 = \frac{1}{C_z}\frac{mg}{\frac{1}{2}\rho S} \tag{17}$$

Replacing the velocity in the previous equation for ϕ:

$$\phi = -\sqrt{\frac{S}{mg}}\sqrt{\frac{\frac{1}{2}\rho}{C_z}}C_{Y_\beta}V_w \tag{18}$$

At a given altitude and C_z, if the wing loading decreases, the bank angle increases for a given side wind. The fast aircrafts, which typically have large wing loading, are therefore less sensitive to crosswinds. However, for a given aircraft, if C_z decreases, that is, if the approach velocity increases, the bank angle increases for a given crosswind.